

Using the Messages API

Practical patterns and examples for using the Messages API effectively

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This guide covers common patterns for working with the Messages API, including basic requests, multi-turn conversations, prefill techniques, and vision capabilities. For complete API specifications, see the [Messages API reference](#).

Basic request and response

Shell ▾



```
#!/bin/sh
curl https://api.anthropic.com/v1/messages \
  --header "x-api-key: $ANTHROPIC_API_KEY" \
  --header "anthropic-version: 2023-06-01" \
  --header "content-type: application/json" \
  --data \
  '{
    "model": "claude-sonnet-4-5",
    "max_tokens": 1024,
    "messages": [
      {"role": "user", "content": "Hello, Claude"}
    ]
  }'
```

JSON



```
"type": "message",
"role": "assistant",
"content": [
  {
    "type": "text",
    "text": "Hello!"
  }
],
"model": "claude-sonnet-4-5",
"stop_reason": "end_turn",
"stop_sequence": null,
"usage": {
  "input_tokens": 12,
  "output_tokens": 6
}
```

Multiple conversational turns

The Messages API is stateless, which means that you always send the full conversational history to the API. You can use this pattern to build up a conversation over time. Earlier conversational turns don't necessarily need to actually originate from Claude — you can use synthetic `assistant` messages.

Shell ▾



```
#!/bin/sh
curl https://api.anthropic.com/v1/messages \
  --header "x-api-key: $ANTHROPIC_API_KEY" \
  --header "anthropic-version: 2023-06-01" \
  --header "content-type: application/json" \
  --data \
'
{
  "model": "claude-sonnet-4-5",
  "max_tokens": 1024,
  "messages": [
    {"role": "user", "content": "Hello, Claude"},
    {"role": "assistant", "content": "Hello!"},
    {"role": "user", "content": "Can you describe LLMs to me?"}
  ]
}'
```

```
{
  "id": "msg_018gCsTGsXkYJVqYPxTgDHBu",
  "type": "message",
  "role": "assistant",
  "content": [
    {
      "type": "text",
      "text": "Sure, I'd be happy to provide..."
    }
  ],
  "stop_reason": "end_turn",
  "stop_sequence": null,
  "usage": {
    "input_tokens": 30,
    "output_tokens": 309
  }
}
```

Putting words in Claude's mouth

You can pre-fill part of Claude's response in the last position of the input messages list. This can be used to shape Claude's response. The example below uses `"max_tokens": 1` to get a single multiple choice answer from Claude.

Shell ▾



```
#!/bin/sh
curl https://api.anthropic.com/v1/messages \
  --header "x-api-key: $ANTHROPIC_API_KEY" \
  --header "anthropic-version: 2023-06-01" \
  --header "content-type: application/json" \
  --data \
'{"
  "model": "claude-sonnet-4-5",
  "max_tokens": 1,
  "messages": [
    {"role": "user", "content": "What is latin for Ant? (A) Apoidea, (B) Rhopalocera,
    {"role": "assistant", "content": "The answer is ("}
  ]
}'
```

```
{
  "id": "msg_01Q8Faay6S7QPTvEuuQARt7h",
  "type": "message",
  "role": "assistant",
  "content": [
    {
      "type": "text",
      "text": "C"
    }
  ],
  "model": "claude-sonnet-4-5",
  "stop_reason": "max_tokens",
  "stop_sequence": null,
  "usage": {
    "input_tokens": 42,
    "output_tokens": 1
  }
}
```

For more information on prefill techniques, see our [prefill guide](#).

Vision

Claude can read both text and images in requests. We support both `base64` and `url` source types for images, and the `image/jpeg`, `image/png`, `image/gif`, and `image/webp` media types. See our [vision guide](#) for more details.

Shell ▾



```
# Option 1: Base64-encoded image
IMAGE_URL="https://upload.wikimedia.org/wikipedia/commons/a/a7/Camponotus_flavomarginatus."
IMAGE_MEDIA_TYPE="image/jpeg"
IMAGE_BASE64=$(curl "$IMAGE_URL" | base64)

curl https://api.anthropic.com/v1/messages \
  --header "x-api-key: $ANTHROPIC_API_KEY" \
  --header "anthropic-version: 2023-06-01" \
  --header "content-type: application/json" \
  --data \
  '{
    "model": "claude-sonnet-4-5",
    "max_tokens": 1024,
    "messages": [
      {
        "role": "user", "content": [
          {
            "type": "image", "source": {
              "type": "base64",
              "media_type": "'$IMAGE_MEDIA_TYPE'",
              "data": "'$IMAGE_BASE64'"
            }
          },
          {
            "type": "text", "text": "What is in the above image?"
          }
        ]
      }
    ]
  }'
```

```
# Option 2: URL-referenced image
curl https://api.anthropic.com/v1/messages \
  --header "x-api-key: $ANTHROPIC_API_KEY" \
  --header "anthropic-version: 2023-06-01" \
  --header "content-type: application/json" \
  --data \
  '{
    "model": "claude-sonnet-4-5",
    "max_tokens": 1024,
    "messages": [
      {
        "role": "user", "content": [
          {
            "type": "image", "source": {
              "type": "url",
              "url": "https://upload.wikimedia.org/wikipedia/commons/a/a7/Camponotus_flavomarginatus."
            }
          },
          {
            "type": "text", "text": "What is in the above image?"
          }
        ]
      }
    ]
  }'
```

```
{
  "id": "msg_01EcyWo6m4hyW8KHs2y2pei5",
  "type": "message",
  "role": "assistant",
  "content": [
    {
      "type": "text",
      "text": "This image shows an ant, specifically a close-up view of an ant. The ant is"
    }
  ],
  "model": "claude-sonnet-4-5",
  "stop_reason": "end_turn",
  "stop_sequence": null,
  "usage": {
    "input_tokens": 1551,
    "output_tokens": 71
  }
}
```

Tool use, JSON mode, and computer use

See our [guide](#) for examples for how to use tools with the Messages API. See our [computer use guide](#) for examples of how to control desktop computer environments with the Messages API.

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